

Technical Requirement ID

TR-INV-001

Related Functional Requirement(s)

FR-DLT-001

Objective

- Implement DLT ingestion of customer CSVs into a managed Delta table.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: customer_csv_source
- Location (Path/Table): /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Format: CSV
- Description: Raw customer files with headers: CustId, Name, EmailId, Region.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Location: Managed table in Unity Catalog
- Format: Delta
- Description: Ingested customer dataset with enforced schema and lineage.

Job Flow / Pipeline Stages

1. Read CSV with header inference and fallback to predefined schema.

2. Normalize column types and write to Delta as a DLT table.

3. Record lineage and operational metrics in DLT event logs.

Data Transformations / Business Logic

- Step:
- Description: Schema enforcement and normalization.
- Transformation Logic: Cast CustId STRING, Name STRING, EmailId STRING, Region STRING; trim whitespace; standardize casing for Region.

Error Handling and Logging

- On schema inference failure, apply predefined schema and log warning; if source empty, create empty table version and emit informational event via DLT expectations metrics.

Technical Requirement ID

TR-INV-002

Related Functional Requirement(s)

FR-DLT-002

Objective

- Implement DLT ingestion of order CSVs into a managed Delta table with malformed record quarantine.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: order_csv_source
- Location (Path/Table): /Volumes/gen_ai_poc_databrickscOE/sdlc_wizard/orderdata

- Format: CSV
- Description: Raw order files with headers: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdmc_wizard.order
- Location: Managed table in Unity Catalog
- Format: Delta
- Description: Ingested orders dataset with error quarantine for malformed rows.

Job Flow / Pipeline Stages

1. Read CSV with schema inference and column mapping.
2. Route malformed records to an error dataset; persist valid rows to Delta.
3. Emit DLT metrics for ingestion counts and quarantined rows.

Data Transformations / Business Logic

- Step:
- Description: Type casting and basic field sanitation.
- Transformation Logic: Cast PricePerUnit DECIMAL(18,2), Qty BIGINT, Date DATE; trim ItemName; standardize OrderId and CustId as STRING.

Error Handling and Logging

- Quarantine malformed rows to gen_ai_poc_databrickscoe.sdmc_wizard.order_error with error reason; log counts in DLT event logs; create empty target version and alert when source path empty.

Technical Requirement ID

TR-INV-003

Related Functional Requirement(s)

FR-DLT-003

Objective

- Compute TotalAmount in orders as PricePerUnit * Qty with type safety.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT

- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: gen_ai_poc_databrickscoe.sdmc_wizard.order
- Location (Path/Table): Unity Catalog table
- Format: Delta
- Description: Ingested orders with typed numeric fields.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdmc_wizard.order_refined
- Location: Managed table in Unity Catalog
- Format: Delta
- Description: Orders with computed TotalAmount and coerced numeric types.

Job Flow / Pipeline Stages

1. Read orders; cast numeric fields with strict semantics.
2. Compute TotalAmount and enforce non-null where inputs valid.
3. Persist refined dataset and publish metrics.

Data Transformations / Business Logic

- Step:
- Description: TotalAmount derivation.
- Transformation Logic: $\text{TotalAmount} = \text{CAST}(\text{PricePerUnit AS DECIMAL}(18,2)) * \text{CAST}(\text{Qty AS DECIMAL}(18,2))$; coerce non-numeric to NULL; flag invalid via expectation.

Error Handling and Logging

- Route rows with non-numeric inputs to a data quality quarantine; record invalid counts in DLT expectations; continue processing valid rows.

Technical Requirement ID

TR-INV-004

Related Functional Requirement(s)

FR-DLT-004

Objective

- Cleanse nulls and deduplicate customer and order datasets using business keys.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: customer, order_refined
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard
- Format: Delta
- Description: Ingested and refined input tables for cleansing.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdlc_wizard.customer_clean, gen_ai_poc_databrickscoe.sdlc_wizard.order_clean
- Location: Managed tables in Unity Catalog
- Format: Delta
- Description: Cleansed datasets without null required keys or duplicates.

Job Flow / Pipeline Stages

1. Normalize textual "Null"/"NULL"/empty to actual nulls.

2. Filter rows with required key nulls per entity.

3. Deduplicate by business key retaining latest by ingestion/event timestamp.

Data Transformations / Business Logic

- Step:
- Description: Null handling and deduplication.
- Transformation Logic: Apply coalesce and nullif on key fields; use window ROW_NUMBER() OVER (PARTITION BY key ORDER BY event_time DESC, ingestion_time DESC) = 1.

Error Handling and Logging

- Log counts of removed null-key rows and duplicates; persist conflict records to _conflict datasets; surface metrics in DLT UI.

Technical Requirement ID

TR-INV-005

Related Functional Requirement(s)

FR-DLT-005

Objective

- Create or validate existence of ordersummary Delta table with defined schema and properties.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: schema_definition_ordersummary
- Location (Path/Table): In-code DLT table definition

- Format: N/A
- Description: Declarative DLT schema for ordersummary including SCD2 metadata columns.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Location: Managed table in Unity Catalog
- Format: Delta
- Description: Target SCD2-capable table with business and metadata columns.

Job Flow / Pipeline Stages

1. Define DLT table with explicit schema and table properties.
2. Apply expectations and enable schema evolution where compatible.
3. Commit creation or evolution with catalog registration.

Data Transformations / Business Logic

- Step:
- Description: Table definition with SCD2 metadata.
- Transformation Logic: Columns include CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date plus IsActive, StartDate, EndDate, RecordHash, SurrogateKey.

Error Handling and Logging

- If existing schema differs, apply additive evolution and log change event; on incompatible change, fail definition and raise alert via DLT logs.

Technical Requirement ID

TR-INV-006

Related Functional Requirement(s)

FR-DLT-006

Objective

- Join cleansed customer and order datasets on CustId to produce enriched order records.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT

- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: customer_clean, order_clean
- Location (Path/Table): gen_ai_poc_databrickscoe.sdmc_wizard
- Format: Delta
- Description: Cleansed customer and order tables ready for join.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdmc_wizard.order_enriched
- Location: Managed table in Unity Catalog
- Format: Delta
- Description: Joined dataset containing customer attributes and order fields including TotalAmount.

Job Flow / Pipeline Stages

1. Perform inner join on CustId.
2. Select required columns for downstream SCD2.
3. Persist enriched dataset and track join metrics.

Data Transformations / Business Logic

- Step:
- Description: Enrichment via join.
- Transformation Logic: Inner join keys: customer_clean.CustId = order_clean.CustId; project Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date, CustId, TotalAmount.

Error Handling and Logging

- Write unmatched order CustId rows to gen_ai_poc_databrickscoe.sdmc_wizard.order_customer_mismatch; log mismatch counts in DLT metrics.

Technical Requirement ID

TR-INV-007

Related Functional Requirement(s)

FR-DLT-007

Objective

- Implement SCD Type 2 management for ordersummary based on CustId+OrderId keys.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: order_enriched, ordersummary (current)
- Location (Path/Table): gen_ai_poc_databrickscoe.sdmc_wizard
- Format: Delta
- Description: Incoming enriched facts and existing target table for comparison.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdmc_wizard.ordersummary
- Location: Managed table in Unity Catalog
- Format: Delta
- Description: History-tracked SCD2 table with single active row per CustId+OrderId.

Job Flow / Pipeline Stages

1. Compute RecordHash on non-key attributes (Name, EmailId, Region, ItemName, PricePerUnit, Qty, Date).
2. Compare with current active records; expire changed rows and insert new versions.
3. Maintain IsActive, StartDate, EndDate, SurrogateKey generation.

Data Transformations / Business Logic

- Step:
- Description: SCD2 upsert.
- Transformation Logic: When hash differs for active key, update existing row IsActive=false, EndDate=current_timestamp; insert new row IsActive=true, StartDate=current_timestamp; ensure only one active per key via merge conditions.

Error Handling and Logging

- Detect multiple active rows per key, retain latest StartDate, flag anomaly to scd2_anomalies dataset; log SCD2 inserts/updates/unchanged counts.

Technical Requirement ID

TR-INV-008

Related Functional Requirement(s)

FR-DLT-008

Objective

- Ensure TotalAmount is persisted accurately within ordersummary for active records.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:

- Name: order_enriched (with TotalAmount)
- Location (Path/Table): gen_ai_poc_databrickscoe.sdmc_wizard
- Format: Delta
- Description: Enriched orders including computed TotalAmount.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdmc_wizard.ordersummary
- Location: Managed table in Unity Catalog
- Format: Delta
- Description: SCD2 table containing validated TotalAmount for active versions.

Job Flow / Pipeline Stages

1. Validate TotalAmount non-null and ≥ 0 .
2. Include TotalAmount in SCD2 insert/update payloads.
3. Exclude invalid TotalAmount rows from upsert and route to quarantine.

Data Transformations / Business Logic

- Step:
- Description: TotalAmount validation and carry-through.
- Transformation Logic: Expect TotalAmount IS NOT NULL AND TotalAmount ≥ 0 ; propagate value unchanged into active records during merge.

Error Handling and Logging

- Persist invalid TotalAmount rows to ordersummary_totalamount_error with reason; log counts via expectations; continue processing valid rows.

Technical Requirement ID

TR-INV-009

Related Functional Requirement(s)

FR-DLT-009

Objective

- Create or validate customeraggregatespend Delta table with defined schema and constraints.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: schema_definition_customeraggregatespend
- Location (Path/Table): In-code DLT table definition
- Format: N/A
- Description: Declarative schema for aggregate output.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdmc_wizard.customeraggregatespend
- Location: Managed table in Unity Catalog
- Format: Delta
- Description: Aggregate table with columns Name, TotalAmount, Date and type constraints.

Job Flow / Pipeline Stages

1. Define DLT table schema and constraints.
2. Register in catalog and enforce data types/nullability.
3. Enable schema evolution for additive, compatible changes.

Data Transformations / Business Logic

- Step:
- Description: Table definition enforcement.
- Transformation Logic: Columns Name STRING NOT NULL, TotalAmount DECIMAL(18,2) NOT NULL, Date DATE NOT NULL; set table properties for quality expectations.

Error Handling and Logging

- On incompatible existing schema, fail deployment and alert; record table creation/evolution events to DLT logs.

Technical Requirement ID

TR-INV-010

Related Functional Requirement(s)

FR-DLT-010

Objective

- Aggregate TotalAmount by Name and Date and load results into customeraggregatespend without duplicates.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: ordersummary (active records)
- Location (Path/Table): gen_ai_poc_databrickscoe.sdmc_wizard
- Format: Delta
- Description: Active SCD2 records containing TotalAmount.

Target Data Details

- Output:
- Name: gen_ai_poc_databrickscoe.sdmc_wizard.customeraggregatespend
- Location: Managed table in Unity Catalog
- Format: Delta
- Description: Daily aggregates of spend per customer name.

Job Flow / Pipeline Stages

1. Filter ordersummary to IsActive = true.
2. Group by Name, Date; sum TotalAmount as TotalAmount.
3. Upsert into customeraggregatespend using merge on Name+Date.

Data Transformations / Business Logic

- Step:
- Description: Deterministic aggregation.
- Transformation Logic: SELECT Name, Date, SUM(TotalAmount) AS TotalAmount FROM active GROUP BY Name, Date; MERGE ON Name AND Date with update on match, insert on not matched.

Error Handling and Logging

- If no source rows, skip write and emit informational metric; validate uniqueness of Name+Date and log duplicates before merge.

Technical Requirement ID

TR-INV-011

Related Functional Requirement(s)

FR-DLT-011

Objective

- Orchestrate the end-to-end pipeline using a single DLT pipeline with configuration and monitoring.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: pipeline_settings
- Location (Path/Table): DLT pipeline configuration
- Format: JSON
- Description: Target catalog/schema, storage location, mode, and triggers.

Target Data Details

- Output:
- Name: All DLT-managed tables and views
- Location: gen_ai_poc_databrickscoe.sdlc_wizard
- Format: Delta
- Description: Persisted datasets across ingestion, refinement, SCD2, and aggregates.

Job Flow / Pipeline Stages

1. Define pipeline with target catalog/schema and storage, set continuous or scheduled mode.
2. Implement datasets with expectations and lineage.
3. Configure triggers, retries, and auto-recovery; deploy and monitor via DLT UI.

Data Transformations / Business Logic

- Step:
- Description: Orchestration rules.
- Transformation Logic: Dataset dependencies defined via DLT decorators; quality expectations enforce pass/drop/quarantine; lineage auto-captured.

Error Handling and Logging

- Enable checkpoint restore and retries; on failure emit alerts and persist event logs; capture run status and metrics for each dataset.

Technical Requirement ID

TR-INV-012

Related Functional Requirement(s)

FR-DLT-012

Objective

- Enforce data quality expectations for schema, nullability, and value ranges across datasets.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: all_intermediate_datasets
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard
- Format: Delta/streaming as applicable
- Description: Datasets subject to expectations.

Target Data Details

- Output:
- Name: quality_metrics, quarantine_datasets
- Location: Managed under catalog/schema
- Format: Delta
- Description: Persisted metrics and quarantined rows for diagnostics.

Job Flow / Pipeline Stages

1. Define expectations for required fields and data types.
2. Apply non-negativity checks for numeric fields.
3. Route failures to quarantine and expose metrics in DLT UI.

Data Transformations / Business Logic

- Step:
- Description: Expectation rules.
- Transformation Logic: Enforce NOT NULL on CustId, OrderId, Date; numeric types for PricePerUnit, Qty, TotalAmount; TotalAmount >= 0; configure onFail actions (drop/quarantine/fail).

Error Handling and Logging

- If failure rate exceeds threshold, fail run and alert; persist error samples and counts for each dataset with timestamps.

Technical Requirement ID

TR-INV-013

Related Functional Requirement(s)

FR-DLT-013

Objective

- Implement incremental processing with idempotent outputs using checkpoints and deterministic merges.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: streaming_csv_sources
- Location (Path/Table): volume paths for customerdata and orderdata
- Format: CSV (incremental)
- Description: New/changed files arriving over time.

Target Data Details

- Output:
- Name: idempotent_targets
- Location: gen_ai_poc_databrickscoe.sdmc_wizard
- Format: Delta
- Description: Tables updated exactly once per file with consistent merges.

Job Flow / Pipeline Stages

1. Enable Auto Loader or incremental reads with checkpoints.
2. Maintain per-source checkpoints to avoid reprocessing.
3. Use deterministic merge conditions for SCD2 and aggregates.

Data Transformations / Business Logic

- Step:
- Description: Incremental semantics.
- Transformation Logic: Source ingestion keyed by file metadata and event-time; merges keyed by business keys; no-op on unchanged hashes.

Error Handling and Logging

- Support backfill toggle to reprocess with controlled overwrite; log checkpoint states; on duplicate detection, skip and record metric.

Technical Requirement ID

TR-INV-014

Related Functional Requirement(s)

FR-DLT-014

Objective

- Apply governance, security, and access control via Unity Catalog.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: permissions_config
- Location (Path/Table): UC grants in deployment scripts
- Format: YAML/SQL
- Description: Role-to-privilege mappings.

Target Data Details

- Output:
- Name: access_controlled_tables
- Location: gen_ai_poc_databrickscoe.sdlc_wizard
- Format: Delta
- Description: Consumer tables with enforced SELECT grants and restricted writes.

Job Flow / Pipeline Stages

1. Grant SELECT on ordersummary and customeraggregatespend to designated roles.
2. Restrict MODIFY/WRITE to DLT service principal.
3. Enable lineage and audit logging in Unity Catalog.

Data Transformations / Business Logic

- Step:
- Description: Security policy application.
- Transformation Logic: Execute GRANT statements and set table properties for lineage/audit; verify privileges post-apply.

Error Handling and Logging

- On grant failure, log event, roll back partial changes, and notify governance group; audit entries captured for all changes.

Technical Requirement ID

TR-INV-015

Related Functional Requirement(s)

FR-DLT-015

Objective

- Provide monitoring, alerting, and logging for pipeline health and data quality.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: dlt_event_logs, expectations_metrics
- Location (Path/Table): System tables and workspace logs
- Format: Delta/JSON
- Description: Operational and quality telemetry.

Target Data Details

- Output:
- Name: monitoring_views_and_alerts
- Location: Workspace dashboards and alerting endpoints
- Format: Views/Notifications
- Description: Surfaces for run status and quality KPIs.

Job Flow / Pipeline Stages

1. Capture DLT event logs and metrics.
2. Configure alerts for failures, schema changes, and quality threshold breaches.
3. Publish run status dashboards.

Data Transformations / Business Logic

- Step:
- Description: Telemetry aggregation.
- Transformation Logic: Aggregate failure counts, durations, and expectation pass rates; compute SLIs for alerting thresholds.

Error Handling and Logging

- If alert delivery fails, retry with exponential backoff; log to fallback sink and record incident reference.

Technical Requirement ID

TR-INV-016

Related Functional Requirement(s)

FR-DLT-016

Objective

- Document datasets, columns, and business rules with Unity Catalog comments and tags.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: metadata_annotations
- Location (Path/Table): UC comments/tags
- Format: Catalog metadata
- Description: Descriptions and governance tags for objects.

Target Data Details

- Output:
- Name: documented_assets
- Location: gen_ai_poc_databrickscoe.sdmc_wizard
- Format: Metadata
- Description: Tables with populated comments and PII tags.

Job Flow / Pipeline Stages

1. Apply table and column comments for all datasets.
2. Tag PII fields such as EmailId.
3. Publish data dictionary in workspace documentation.

Data Transformations / Business Logic

- Step:
- Description: Metadata enrichment.
- Transformation Logic: Use ALTER TABLE/ALTER COLUMN SET COMMENT; APPLY TAG pii=true ON COLUMN EmailId.

Error Handling and Logging

- If tagging restricted, log denial and notify governance; track documentation completion status.

Technical Requirement ID

TR-INV-017

Related Functional Requirement(s)

FR-DLT-017

Objective

- Implement tests validating transformations, SCD2 behavior, and aggregates with CI evidence.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:

- Name: test_input_datasets
- Location (Path/Table): Dedicated test schema/paths
- Format: Delta/CSV
- Description: Known-outcome datasets for unit and integration tests.

Target Data Details

- Output:
- Name: test_results_evidence
- Location: CI logs and test schema
- Format: Delta/Logs
- Description: Stored assertions and pass/fail outcomes.

Job Flow / Pipeline Stages

1. Execute unit tests for TotalAmount, null handling, deduplication.
2. Validate SCD2 state transitions on controlled changes.
3. Verify aggregate sums for specific dates.

Data Transformations / Business Logic

- Step:
- Description: Assertion logic.
- Transformation Logic: Compare computed columns to expected values; assert single active per key; compare aggregate totals to golden records.

Error Handling and Logging

- On test failure, block promotion and emit CI artifacts; create defect tickets with repro data.

Technical Requirement ID

TR-INV-018

Related Functional Requirement(s)

FR-DLT-018

Objective

- Configure scheduling and SLAs for pipeline freshness and availability.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core

- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: scheduling_policy
- Location (Path/Table): Pipeline configuration
- Format: JSON
- Description: Mode and schedule definitions.

Target Data Details

- Output:
- Name: sla_tracking_records
- Location: Monitoring store
- Format: Delta
- Description: SLA adherence metrics and breach indicators.

Job Flow / Pipeline Stages

1. Set pipeline to continuous or scheduled batch per SLA.
2. Define freshness targets for ordersummary and customeraggregatespend.
3. Track run completion vs. targets.

Data Transformations / Business Logic

- Step:
- Description: SLA evaluation.
- Transformation Logic: Compute latency between data arrival and table availability; mark pass/fail per run.

Error Handling and Logging

- On SLA breach, trigger incident workflow and log breach details; retain history for trend analysis.

Technical Requirement ID

TR-INV-019

Related Functional Requirement(s)

FR-DLT-019

Objective

- Implement error persistence and parameterized reprocessing for bad data.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: error_quarantine_stores
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard
- Format: Delta
- Description: Persisted error rows with reasons.

Target Data Details

- Output:
- Name: reprocess_job_output
- Location: Managed targets after replay
- Format: Delta
- Description: Corrected datasets after parameterized reprocessing.

Job Flow / Pipeline Stages

1. Persist quarantined rows with error metadata.

2. Provide reprocess parameter (date/file/key) to replay impacted data.
3. Re-ingest and merge corrected rows deterministically.

Data Transformations / Business Logic

- Step:
- Description: Targeted replay.
- Transformation Logic: Select errors by parameter, cleanse/fix, re-run through transformations; idempotent merges ensure consistency.

Error Handling and Logging

- On repeated failures, capture diagnostics and escalate; maintain reprocess audit trail with timestamps and operator IDs.

Technical Requirement ID

TR-INV-020

Related Functional Requirement(s)

FR-DLT-020

Objective

- Ensure scalable performance via partitioning, optimization, and autoscaling.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: performance_inputs
- Location (Path/Table): All operational tables

- Format: Delta
- Description: Data subject to tuning.

Target Data Details

- Output:
- Name: optimized_tables
- Location: gen_ai_poc_databrickscoe.sdlc_wizard
- Format: Delta
- Description: Partitioned and optimized tables with statistics.

Job Flow / Pipeline Stages

1. Partition ordersummary and aggregates by Date.
2. Apply OPTIMIZE/ZORDER and ANALYZE TABLE for key columns.
3. Configure autoscaling and adjust shuffle/parallelism settings.

Data Transformations / Business Logic

- Step:
- Description: Physical optimization.
- Transformation Logic: Z-Order on CustId, OrderId for ordersummary; collect stats for join/merge predicates; cache dimension where beneficial.

Error Handling and Logging

- If runtime exceeds threshold, emit alert and log performance metrics; capture before/after optimization deltas.

Technical Requirement ID

TR-INV-021

Related Functional Requirement(s)

FR-DLT-021

Objective

- Support backfill and historical loads with correct SCD2 semantics.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT

- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: historical_files
- Location (Path/Table): Source volume partitions
- Format: CSV
- Description: Backdated customer and order data.

Target Data Details

- Output:
- Name: consistent_historical_targets
- Location: gen_ai_poc_databrickscoe.sdmc_wizard
- Format: Delta
- Description: Targets populated with accurate historical lineage and aggregates.

Job Flow / Pipeline Stages

1. Enable backfill mode parameter to select historical partitions.
2. Process in event-time order for correctness.
3. Apply SCD2 with StartDate/EndDate aligned to event-time when available.

Data Transformations / Business Logic

- Step:
- Description: Time-aware processing.
- Transformation Logic: When event-time missing, use ingestion-time; document limitation; maintain determinism in merges.

Error Handling and Logging

- On anomalies during backfill, log and quarantine affected partitions; track backfill run metadata and outcomes.

Technical Requirement ID

TR-INV-022

Related Functional Requirement(s)

FR-DLT-022

Objective

- Manage schema enforcement and controlled evolution across datasets.

Target Cluster Configuration

- Cluster Name: dlt-inventory-core
- Databricks Runtime Version: 13.3.x LTS (Photon) with DLT
- Node Type: i3.xlarge equivalent (DBR-supported)
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Enabled (2–8 workers)
- Auto Termination: 30 minutes idle
- Libraries Installed:
- com.databricks:libraries-dlt-core

Source Data Details

- Dataset:
- Name: schema_definitions_registry
- Location (Path/Table): Repository/config
- Format: JSON/YAML
- Description: Versioned schemas for customer and order.

Target Data Details

- Output:
- Name: schema_versioned_tables
- Location: gen_ai_poc_databrickscoe.sdmc_wizard
- Format: Delta
- Description: Tables with enforced schemas and tracked versions.

Job Flow / Pipeline Stages

1. Enforce expected schemas on ingest.

2. Allow additive evolution with approval workflow.
3. Version and communicate schema changes to consumers.

Data Transformations / Business Logic

- Step:
- Description: Schema controls.
- Transformation Logic: Enable schema enforcement and evolution flags; maintain schema_version column/property; validate compatibility pre-deploy.

Error Handling and Logging

- On incompatible change, fail run, emit alert, and record diff; maintain audit of schema versions and approvals.